Linear Discriminanat Analysis

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Introduction The aim of this assignment is to get familiar with Linear Discriminant Analysis (LDA). LDA and Principal Component Analysis (PCA) are two techniques for dimensionality reduction. PCA can be described as an unsupervised algorithm that ignores data labels and aims to find directions which maximalize the variance in a data. In comparison with PCA, LDA is a supervised algorithm and aims to project a dataset onto a lower dimensional space with good class separability. In other words, LDA maximalizes the ratio of betweenclass variance and the within-class variance in a given data.

Input data In this tutorial, we will work with a dataset that classifies wines (samples) into three classes using of 13 continuous attributes; for more details see wine info.txt file. The dataset is located at wine.csv.

Linear Discriminant Analysis As we mentioned above, LDA finds directions where classes are well-separated, i.e. LDA maximizes the ratio of between-class variance and the within-class variance. Firstly, assume that C is a set of classes and set D, which represents a training dataset, is defined as $D = \{x_1, x_2, ..., x_N\}$.

The between-classes scatter matrix SB is defined as: $S_b = \sum_c N_C(\mu_c - \overline{x})(\mu_c - \overline{x})^T$, where $\overline{x}$ is a vector represents the overall mean of the data, µ represents the mean corresponding to each class, and N_C are sizes of the respective classes.

The within-classes scatter matrix S_W is defined as:

$S_W = \sum_c \sum_{x \in D_c} (x - \overline{\mu_c})(x - \overline{\mu_c})^T$

Next, we will solve the generalized eigenvalue problem for the matrix $S_W^{-1} S_B$ to obtain the linear discriminants, i.e.

$(S_W^{-1} S_B)w = \lambda w$

where w represents an eigenvector and λ represents an eigenvalue. Finally, choose k eigenvectors with the largest eigenvalue and transform the samples onto the new subspace.

Step by step

```r
library(dplyr)
```

Load the dataset

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
wine <- read.csv("wine.csv", header=FALSE)
names(wine) <- c("class", "alcohol", "acid", "ash", "alcalinity", "mg", "phenols", "flavanoids","nf-phe
```

```r
ComputeWithinScatter <- function(data, n)
{
  withinMatrix = matrix(0, ncol(data[[1]]), ncol(data[[1]]))
  for (i in 1:n) {
    group <- data[[i]]
    group.mean <- colMeans(as.matrix(group))

    for (j in 1:nrow(group)) {
      inner <- t(as.matrix(group[j,])) - group.mean
      res <- inner %*% t(inner)

      withinMatrix <- withinMatrix + res
    }
  }
  return(withinMatrix)
}
```

Compute the within-scatter matrix

```r
ComputeBetweenScatter <- function(data, n, meanOverall)
{
  betweenMatrix = matrix(0, ncol(data[[1]]), ncol(data[[1]]))
  for (i in 1:n) {
    group <- data[[i]]
    group.mean <- colMeans(as.matrix(group))

    inner <- group.mean - meanOverall
    res <- nrow(group) * inner %*% t(inner)

    betweenMatrix <- betweenMatrix + res
  }
  return(betweenMatrix)
}
```

Compute the between-scatter matrix

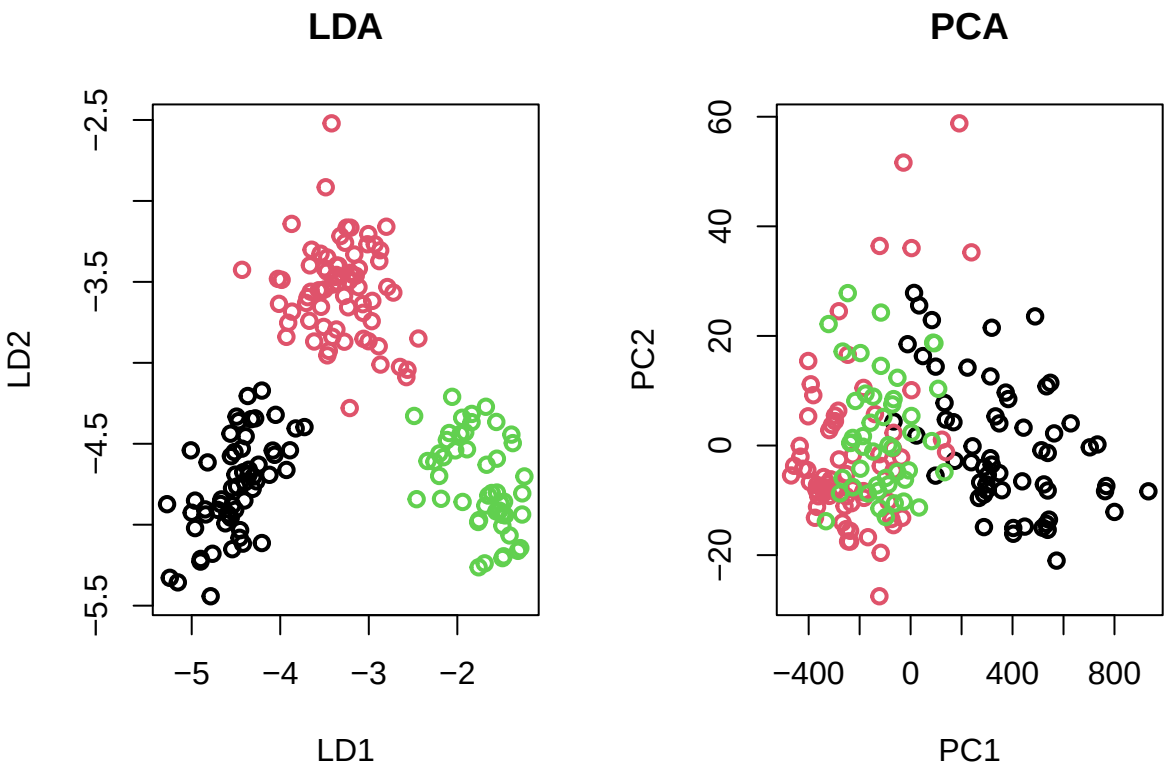
```r
SolveEigenProblem <- function(withinMatrix, betweenMatrix, prior)
{
  Sw_inv <- solve(withinMatrix)
  A <- Sw_inv %*% betweenMatrix

  eivectors <- eigen(A)

  return(eivectors)
}
```

Solve the EigenProblem and return eigen-vector

Visualize the results Project your data into lower-dimensional subspace, visualize this projection, and compare with PCA (see Fig. 1). Also, try to use scale/unscale version of `prcomp` function in R. Use the following code while filling in the lines marked as `TODO`.

```
ComputeCentroids <- function(data, labels){
  yGroupedMean <- aggregate(as.data.frame(data), by = list(labels), FUN = mean)
  rownames(yGroupedMean) <- yGroupedMean[,1]
  yGroupedMean <- yGroupedMean[,-1]
  return(yGroupedMean)
}

Classify <- function(newData, eigenVectors, labels, centroids){
  y <- as.matrix(newData) %*% eigenVectors[,1:(length(levels(labels))-1)]
  prior <- table(labels)/sum(table(labels))

  classification <- matrix(nrow = nrow(newData), ncol = length(levels(labels)))
  colnames(classification) <- levels(labels)
  for(c in levels(labels))
  {
    classification[,c] <- as.matrix(0.5*rowSums((y - matrix(rep(as.matrix(centroids[c,]),
                                                                  nrow(newData)), nrow = nrow(newData),
                                                          byrow = TRUE) )^2)
                                    - log(prior[c]))
  }
  return(levels(labels)[apply(classification, MARGIN = 1, which.min)])
}

CrossvalidationLDA <- function(mydata, labels, kfolds = 10){
  set.seed(17)
  #randomly shuffle the data
  random <- sample(nrow(mydata))
  data <-mydata[random,]
  labels <- labels[random]
  #Create 10 equally size folds
  folds <- cut(seq(1,nrow(data)),breaks=kfolds,labels=FALSE)
  acc <- rep(0, times = kfolds)
  #10 fold cross validation
  for(i in 1:kfolds){
    #Segment your data by fold using the which() function
    testIndexes <- which(folds==i,arr.ind=TRUE)
    testData <- data[testIndexes, ]
    trainData <- data[-testIndexes, ]
    testLabels <- labels[testIndexes]
    trainLabels <- labels[-testIndexes]

    eigenLDA <- LDA(trainData, trainLabels)
    centroids <- ComputeCentroids(as.matrix(trainData) %*% eigenLDA[,1:(length(levels(trainLabels))-1)]
                                  labels = trainLabels)
    pre <- Classify(newData = testData, labels = trainLabels, eigenVectors = eigenLDA,
                    centroids = centroids)
    acc[i] <- sum(pre == testLabels)/length(testLabels)
  }
  return(mean(acc))
}
```

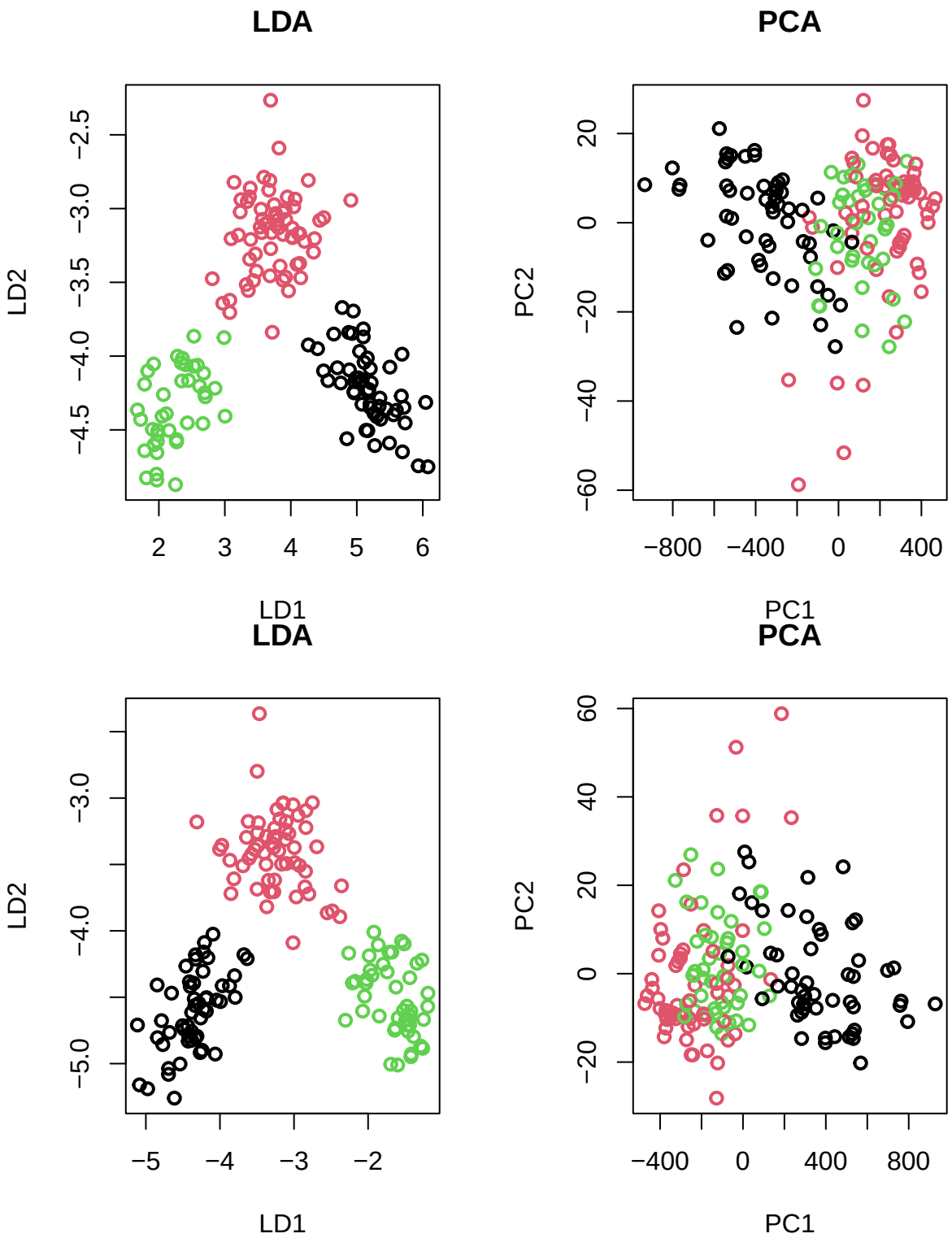
```r
LDA <- function(mydata, labels){

  #number of classes
  n <-length(levels(labels))

  # 1) split the data w.r.t. given factors
  splittedData <- split(mydata, labels)

  # 2) scatter matrices
  #############  within-class scatter matrix Sw ##################
  withinScatterMatrix <- ComputeWithinScatter(splittedData, n) #TODO

  #############  between-class scatter matrix Sb ##################
  meanOverall = colMeans(as.matrix(mydata))
  betweenScatterMatrix <- ComputeBetweenScatter(splittedData, n, meanOverall) #TODO

  # 3)  eigen problem
  ############ solve Eigen problem ################################
  ei <- SolveEigenProblem(withinScatterMatrix, betweenScatterMatrix)

  #transform the samples onto the new subspace
  y <- (as.matrix(mydata) %*% ei$vectors[,1:2])

  ## visual comparison with PCA
  par(mfrow=c(1,2))
  pca <- prcomp(mydata)
  plot(y[,1], y[,2], col = labels, pch = 21, lwd = 2, xlab = "LD1" , ylab = "LD2", main = "LDA")
  plot(-pca$x, col = labels, pch = 21, lwd = 2, main = "PCA")

  return(ei$vectors)
}

############################## FUNCTIONS END ####################################


############################## MAIN ##########################################

### PREPARE DATA
#data(iris)
#mydata <- iris
#labels <- mydata[,5]
#mydata <- mydata[,-5]

mydata <- read.csv("wine.csv", header = FALSE)
labels <- mydata[,1]
labels <- as.factor(labels)
mydata <- mydata[,-1]

#compute LDA and return corresponding eigenvectors
eigenLDA <- LDA(mydata, labels)
```

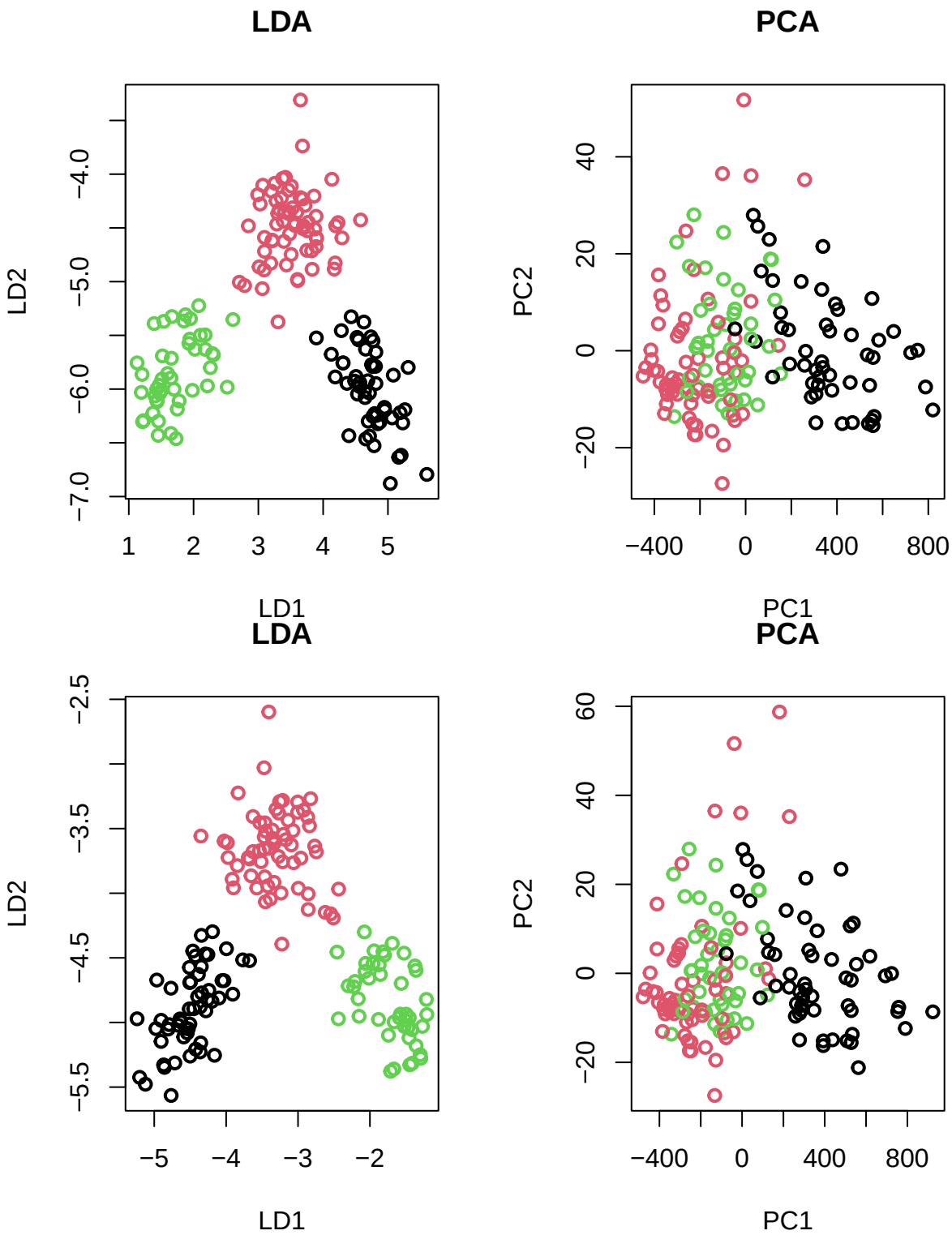


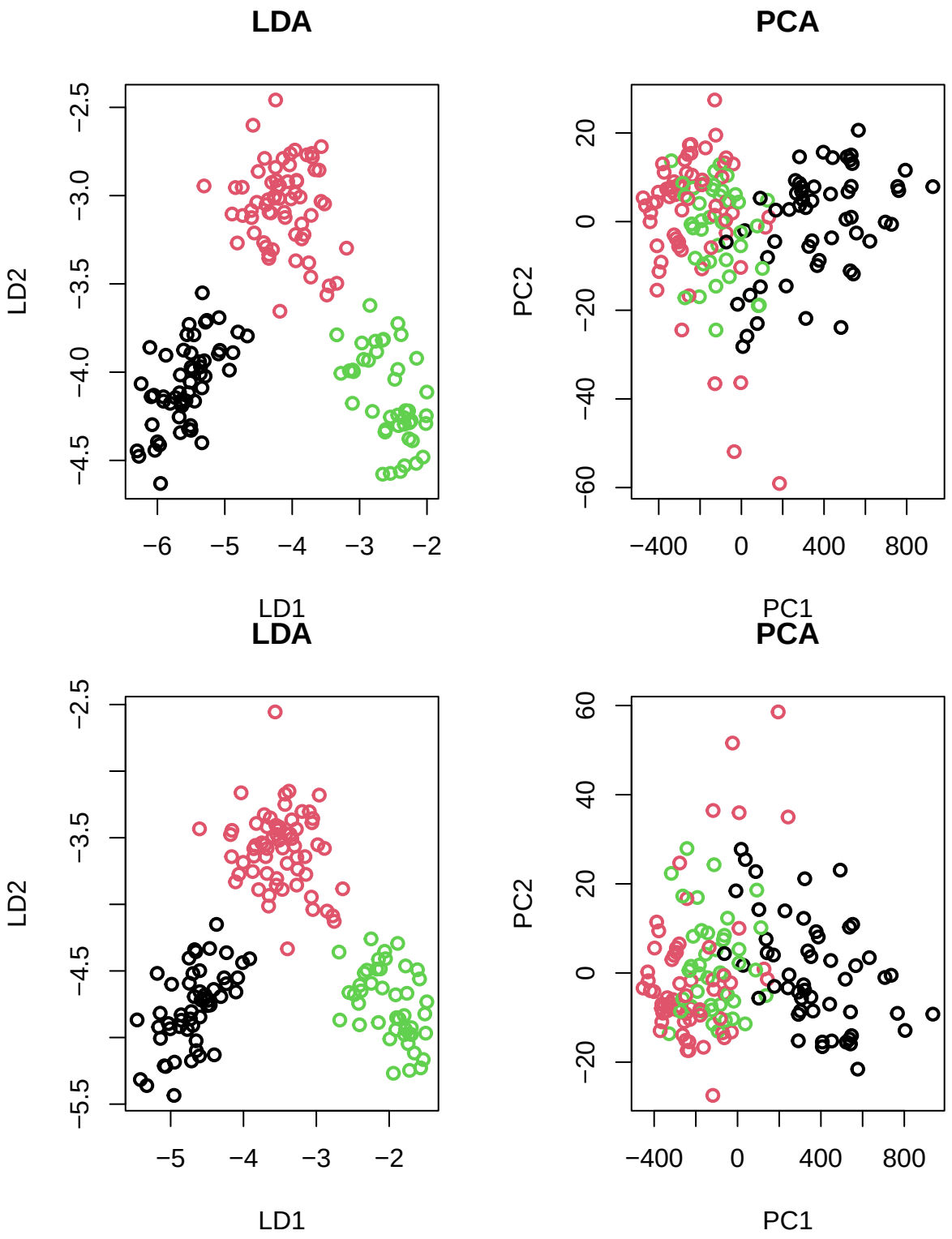

```
#find centroids in the transformed data
centroids <- ComputeCentroids(as.matrix(mydata) %*% eigenLDA[,1:(length(levels(labels))-1)],
                              labels = labels)
#make predictions on the "mydata"
prediction <- Classify(newData = mydata, labels = labels, eigenVectors = eigenLDA,
         centroids = centroids)
#ACC
sum(prediction == labels)/(length(labels))
```

```
## [1] 0.988764
```

```
#CrossValidation
accLDA <- CrossvalidationLDA(mydata, labels, kfolds = 10)
```


LDA
LD2
LD1
−2.5
−3.0
−3.5
−4.0
−4.5
2
3
4
5
6
PCA
PC2
PC1
20
0
−20
−40
−60
−800
−400
0
400
LDA
LD2
LD1
−3.0
−4.0
−5.0
−5
−4
−3
−2
PCA
PC2
PC1
60
40
20
0
−20
−400
0
400
800


LDA
LD2
−4.0
−5.0
−6.0
−7.0
1
2
3
4
5
LD1
PCA
PC2
40
20
0
−20
−400
0
400
800
PC1
LDA
LD2
−2.5
−3.5
−4.5
−5.5
−5
−4
−3
−2
LD1
PCA
PC2
60
40
20
0
−20
−400
0
400
800
PC1


LDA
LD2
LD1
−2.5
−3.0
−3.5
−4.0
−4.5
−6
−5
−4
−3
−2
PCA
PC2
PC1
20
0
−20
−40
−60
−400
0
400
800
LDA
LD2
LD1
−2.5
−3.5
−4.5
−5.5
−5
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−3
−2
PCA
PC2
PC1
60
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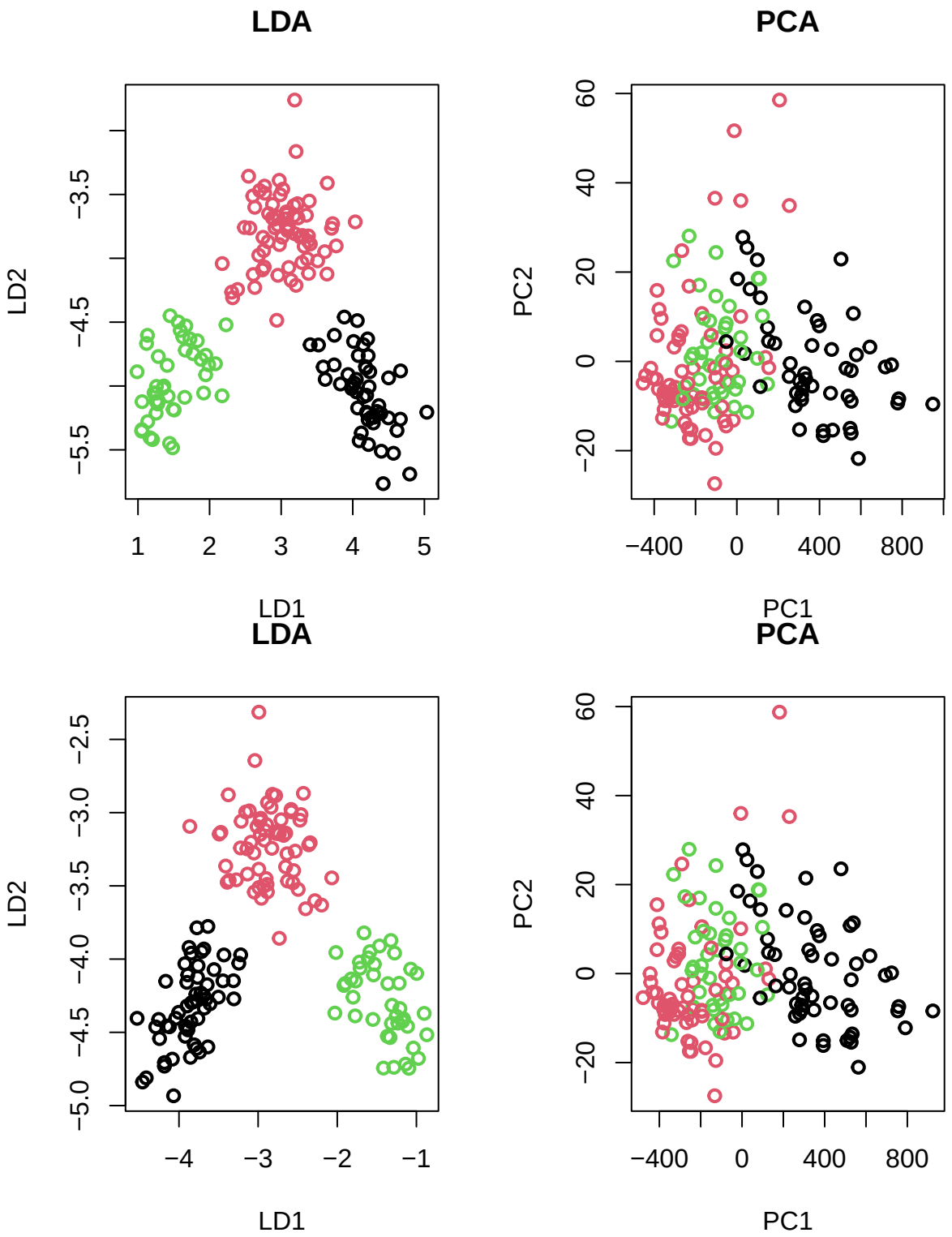
LDA
LD2
LD1
−3.5
−4.5
−5.5
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5
PCA
PC2
PC1
60
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−20
−400
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LDA
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−1
PCA
PC2
PC1
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−20
−400
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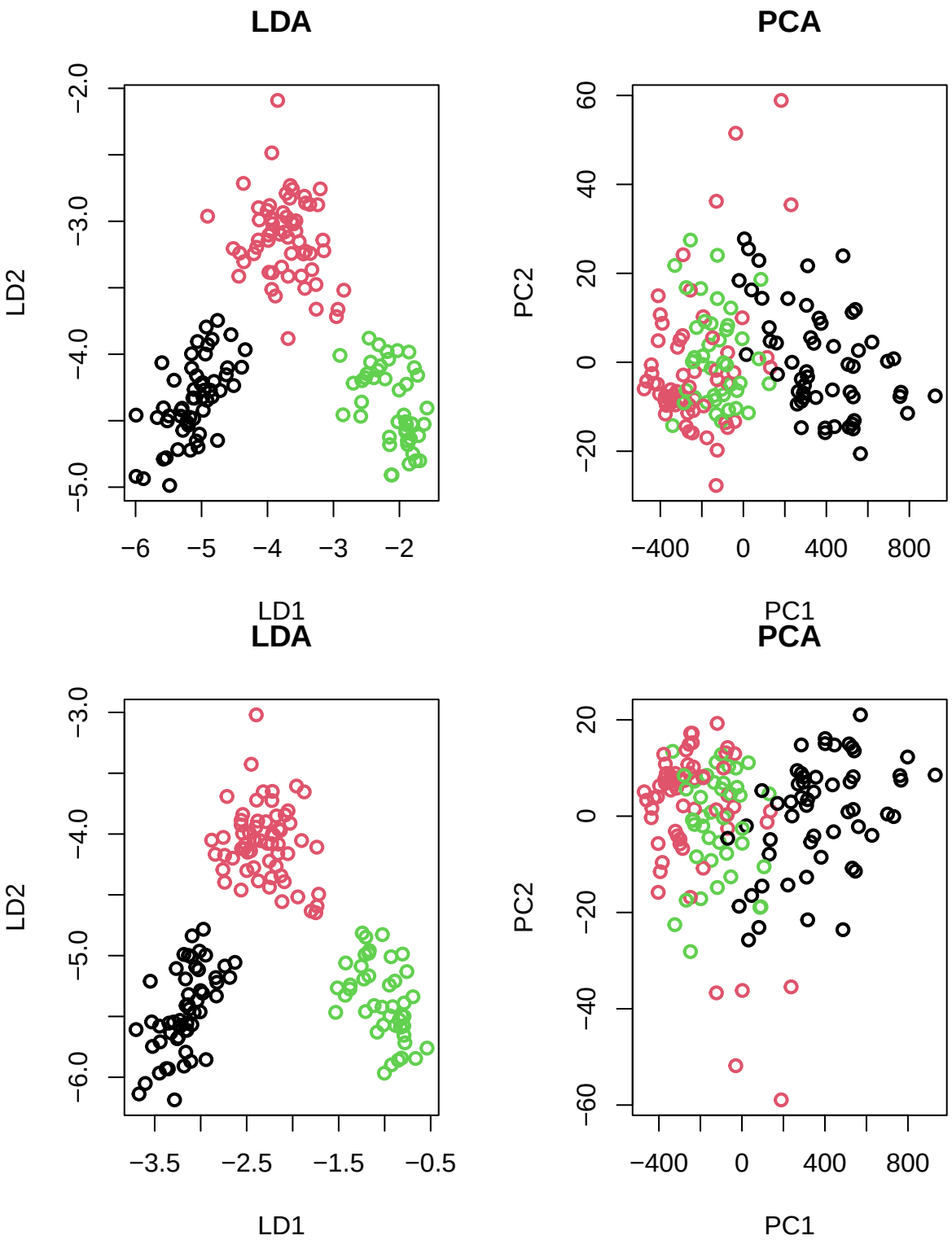
800



Discuss given results. Crossvalidation sugests, that our accuracy is 99.8%. This suggests, that for our data, LDA is a strong method. This can be confirmed visualy, where PCA fairly often mixes data.